PIPELINE AUDIT AND HEALTH REPORT

Semiconductor: re-entrant wafer fab

A wafer lot passes lithography, etch, deposition and metrology, then loops back to lithography for its next layer (62% of the time) until its layers are finished. Every lot visits the scanner about 2.6 times, so the lithography tool is the constraint, and its breakdowns hurt the most. The loop uses a bypass lane to avoid deadlock.

Generated 2026-10-13 19:48 | simulated 18.0 h (warm-up 60 min) | seed 15 | 10 replications

HEALTH SCORE

86 /100

Grade B

Good: sound design with minor weaknesses worth addressing.

Critical 0 | High 0 | Medium 2 | Low 2 | Info 1

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
4.9 units/h	85 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
5.2 units	3137 s	litho

Executive summary

The line keeps up with demand (0.00% lost, sources never throttled): throughput equals the arrival rate, so nothing is binding right now. The busiest station is 'litho' at 67% busy; that is where capacity runs out first as demand grows.

What to fix first

- Medium (litho): Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.
- Medium (litho, deposition): Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.
- Low (throughput): Use a longer run if you need tighter estimates.
- Low (run length): Aim for at least 30x the cycle time so averages settle.

Best measured improvement

No single local change moves throughput by more than 1%: the line is demand-limited or constrained elsewhere. Try raising the arrival rate, or look at the system-wide structure.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Medium	Reliability	**litho**	Down 15% of the time (11 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Medium	Reliability	**litho, deposition**	Every unit passes through these stations, and they can fail. A single failure stops the whole plant. Action: Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.	-8
Low	Confidence	**throughput**	Throughput 95% CI is +-6.1% of the mean. Action: Use a longer run if you need tighter estimates.	-1
Low	Model	**run length**	The measured window (17.0 h) is only 20x the cycle time. Action: Aim for at least 30x the cycle time so averages settle.	-2
Info	Capacity	**litho**	Highest utilisation in the line (83%); the line currently keeps up with demand. Action: This is where capacity runs out first as demand grows.	

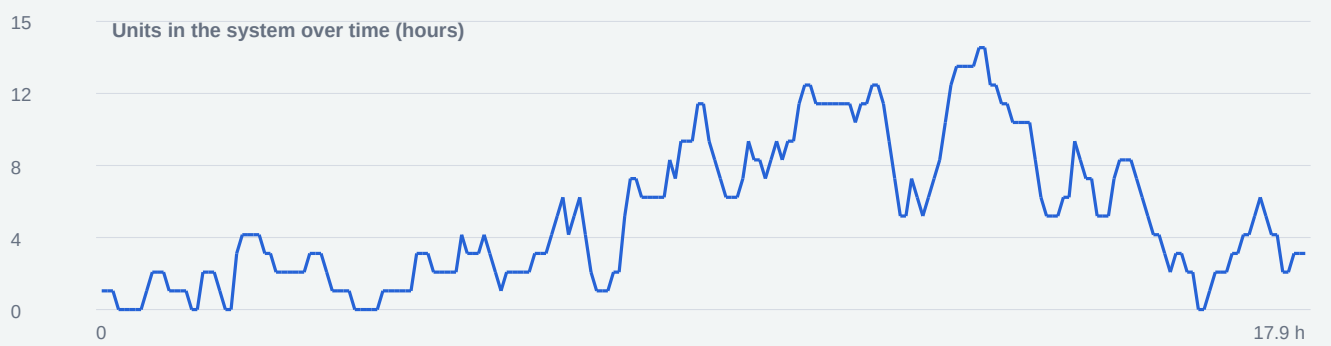
Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
lots	source	-	-	-	-	-
gown	delay		9.1	5.7	0.0	0.2 / 2
litho bottleneck	machine		67.2	0.0	15.4	3.5 / 10
etch	machine		42.4	0.5	0.0	0.4 / 2
deposition	machine		50.6	0.0	5.4	0.9 / 7
metrology	machine		19.3	0.0	0.0	0.2 / 1
wafer_test	sink	-	-	-	-	-
scrap	sink	-	-	-	-	-

Work in process over time



Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	5.02 units/h	+- 0.30	0.43	6.1%
Average WIP	4.91 units	+- 1.26	1.77	25.7%
Cycle time	2941.39 s	+- 690.06	964.70	23.5%
Arrivals lost	0.00 %	+- 0.00	0.00	0.0%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
litho	60 min	10.0 min	85.7%	15.4%	11
deposition	120 min	15.0 min	88.9%	5.4%	4

Single points of failure

All units must pass through: litho, etch, deposition, metrology. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 105 = completed 85 + scrapped 18 + lost 0 + in system 2
Little's Law	PASS	L = 5.22, lambda x W = 5.18 (gap 0.8%)
Run length	WARN	window = 20 x cycle time (30x or more recommended)
Warm-up	PASS	first 60 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 5.0 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
litho	10% faster processing	5.1	**+0.0**	+0.8%
litho	20% shorter repairs	5.1	**+0.0**	+0.4%
litho	+2 buffer slots	5.0	**+0.0**	+0.0%
deposition	+2 buffer slots	5.0	**+0.0**	+0.0%
deposition	20% shorter repairs	5.0	**+0.0**	+0.0%
etch	+2 buffer slots	5.0	**+0.0**	+0.0%
etch	10% faster processing	5.0	**+0.0**	+0.0%
deposition	10% faster processing	5.0	**-0.0**	-0.8%

No single local change moves throughput by more than 1%: the line is demand-limited or constrained elsewhere. Try raising the arrival rate, or look at the system-wide structure.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
lots	source	interarrival=exponential(600), on_blocked=wait
gown	delay	transit=60
litho	machine	capacity=8, mtbf=3600, mttr=600, process=normal(160,20)
etch	machine	capacity=6, process=normal(100,15)
deposition	machine	capacity=6, mtbf=7200, mttr=900, process=normal(120,15)
metrology	machine	capacity=6, process=normal(45,8)
wafer_test	sink	(defaults)
scrap	sink	counts_as=scrapped

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
Medium	**litho**	Down 15% of the time (11 failures).	-3
Medium	**litho, deposition**	Every unit passes through these stations, and they can fail. A single failure stops the whole plant.	-8
Low	**throughput**	Throughput 95% CI is +-6.1% of the mean.	-1
Low	**run length**	The measured window (17.0 h) is only 20x the cycle time.	-2
	Total deductions		**-14**
	Health score		**86**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.